EPFL - Computer Science

CS-401 Applied Data Analysis

Lecture 03 — September 29, 2025

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Visualizing Data

Uses for data visualization

Flow: Ask an interesting question ⇒ Get the data ⇒ Explore the data ⇒ Model the data ⇒ Communicate and visualize the results.

Analysis: Support reasoning about information

- Finding relationships
- Discover structure
- Quantifying values and influences
- Should be part of the data analysis cycle

Communication: Inform and persuade other

- Capture attention, engage
- Tell a story visually
- Can focus on certain aspects and omit other

Decision making: Make it easier to evaluate potential courses of action

Static vs interactive visualization

Static viz: Great for data exploration, developed throughout the last few centuries...

Interactive viz: More and more common when delivering results (and also during exploration). New frameworks are the key enabler.

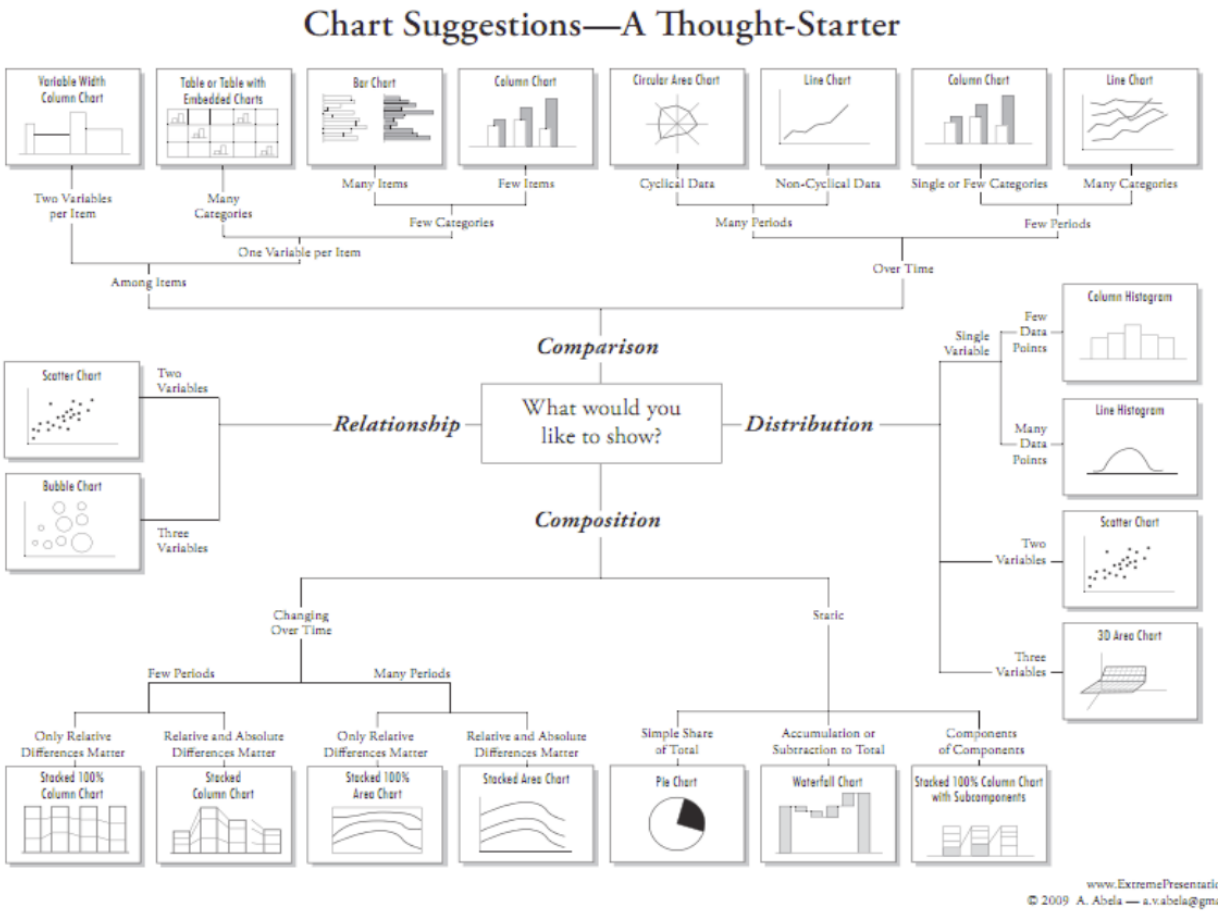
In this course we will focus on **static visualisation**.

There's a dedicated course: COM-480 "Data Visualization"

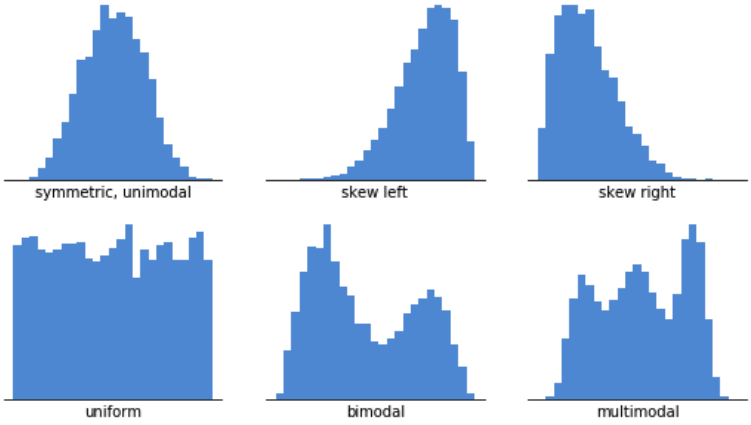
Today's lecture

- Part 1: Navigating the chart landscape
- Part 2: Principles and best practices
- Part 3: A (small) selection of use cases for data visualization

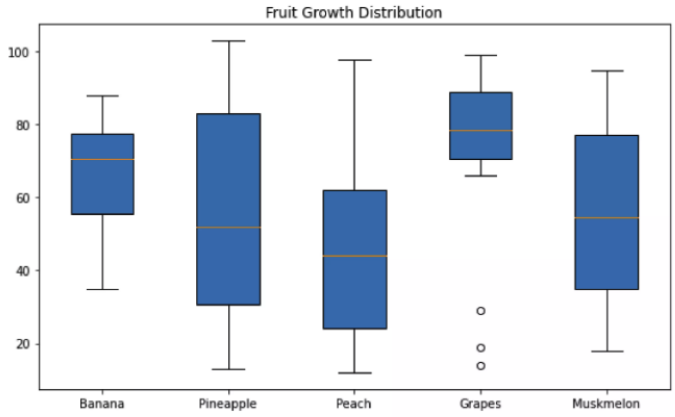
Part 1: Navigating the chart landscape



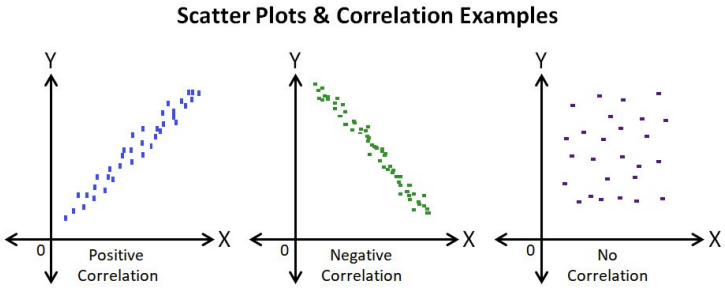
- **One variable**
  - Histograms can tell you a lot about a single variable, discrete or continuous. Easy to recognize skewed distributions! Very powerful data visualization for studying data distribution.

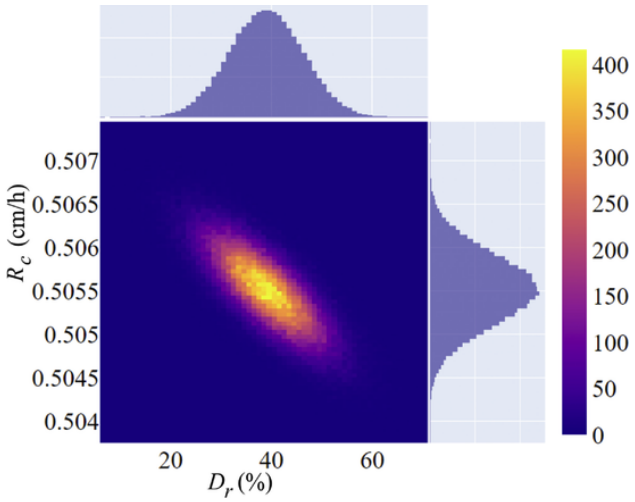


- Box plots is a method for demonstrating graphically the locality, spread and skewness groups of numerical data through their quartiles. You can see from the Median is the data is normalized or skewed distributed.



- **Two variables**
  - Scatter plots quickly expose the relationships between two variables, often exposing correlation between variables. It's useful when there isn't too many data points. If there are two many data points, some solutions are 2D histograms (a.k.a heatmap)

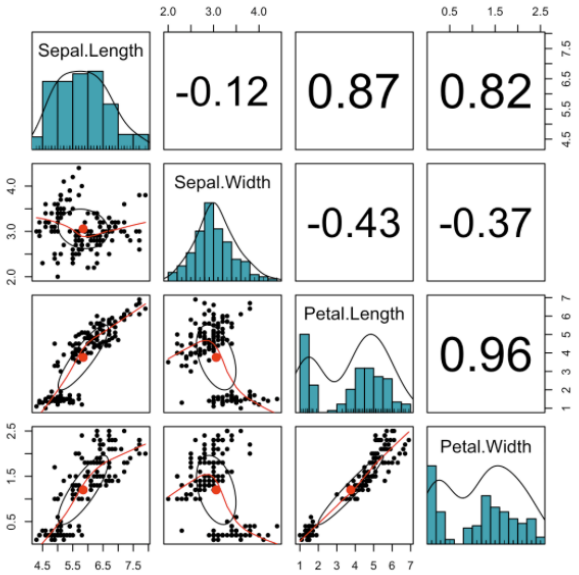




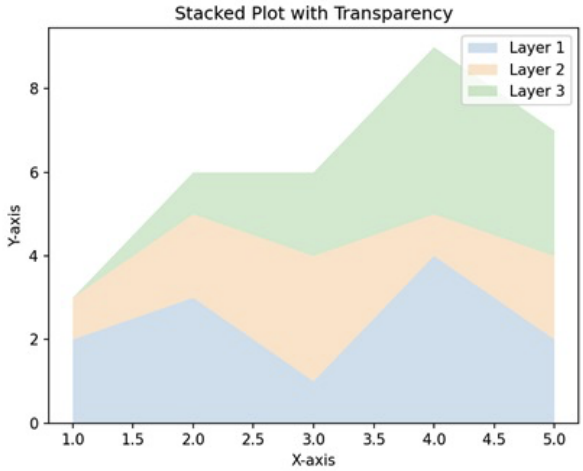
  - Line plots are useful if the relationship is functional (for instance, after binning and aggregating). For instance, days in the x-axis and stock price in the y-axis.



- **> 2 variables**
  - Scatter plot matrix



– Stacked plots often used with many variables: stack index, height, color, etc. Often color variables are categorical, height variables are continuous, stack can be variable and etc.

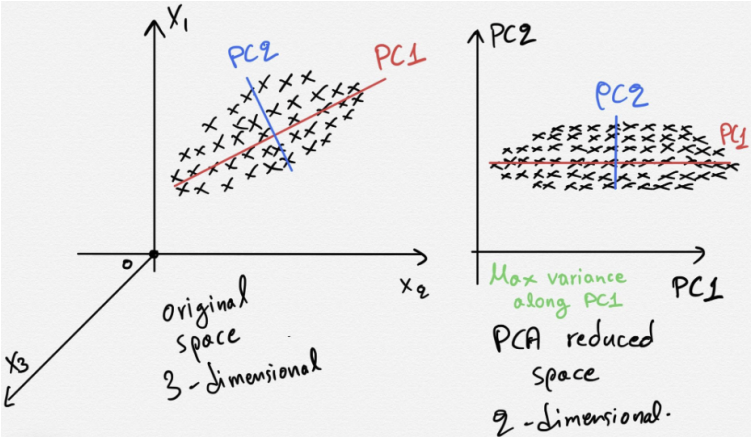


Note

Use the space wisely when you do data visualization to convey useful information rather than redundant information.

Dimensionally reduction

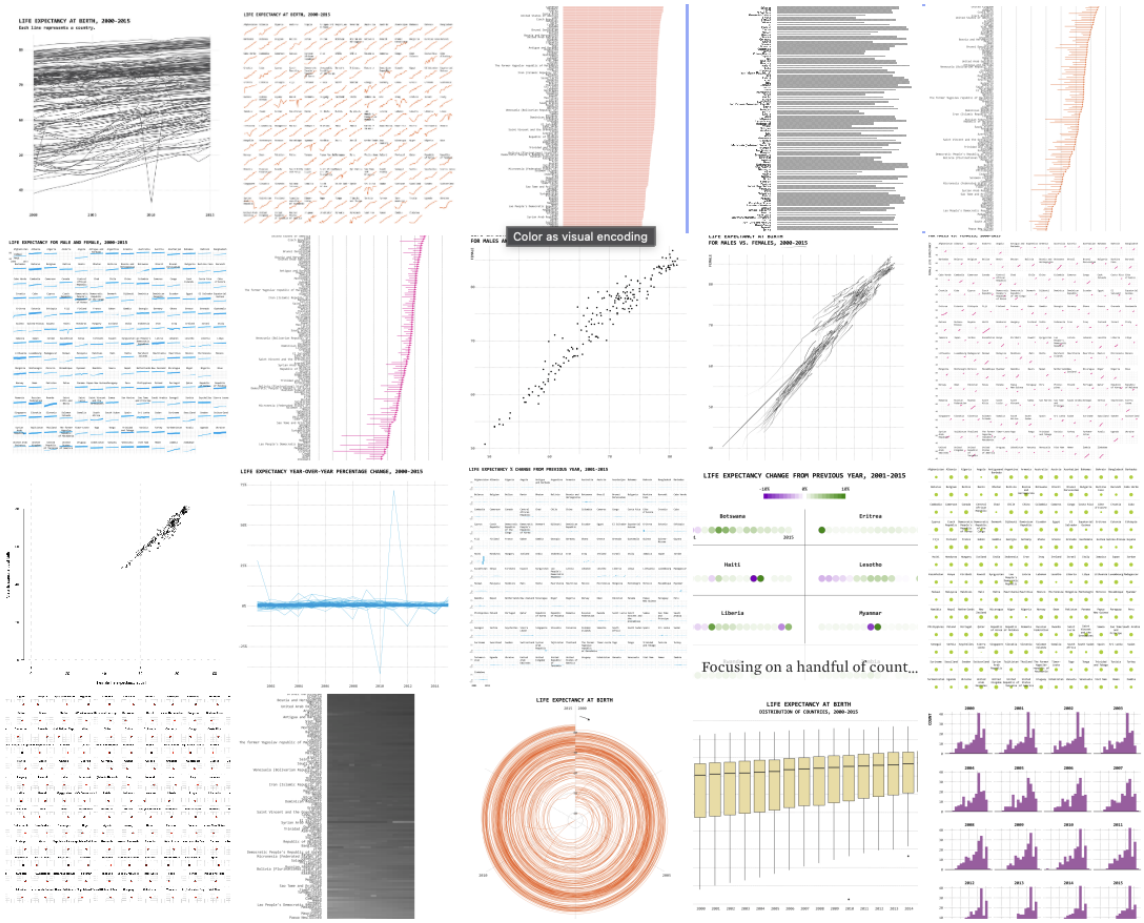
- For example, **PCA**: allows visualization of high-dimensional continuous data in 2D using principal components
- The principal components are the strongest (highest variation) dimensions in the dataset, and are orthogonal.



Important example: one dataset, visualized 25 ways

http://flowingdata.com/2017/01/24/one-dataset-visualized-25-ways

"You must help the data focus and get to the point. Otherwise, it just ends up rambling about what it had for breakfast this morning and how the coffee was not hot enough."



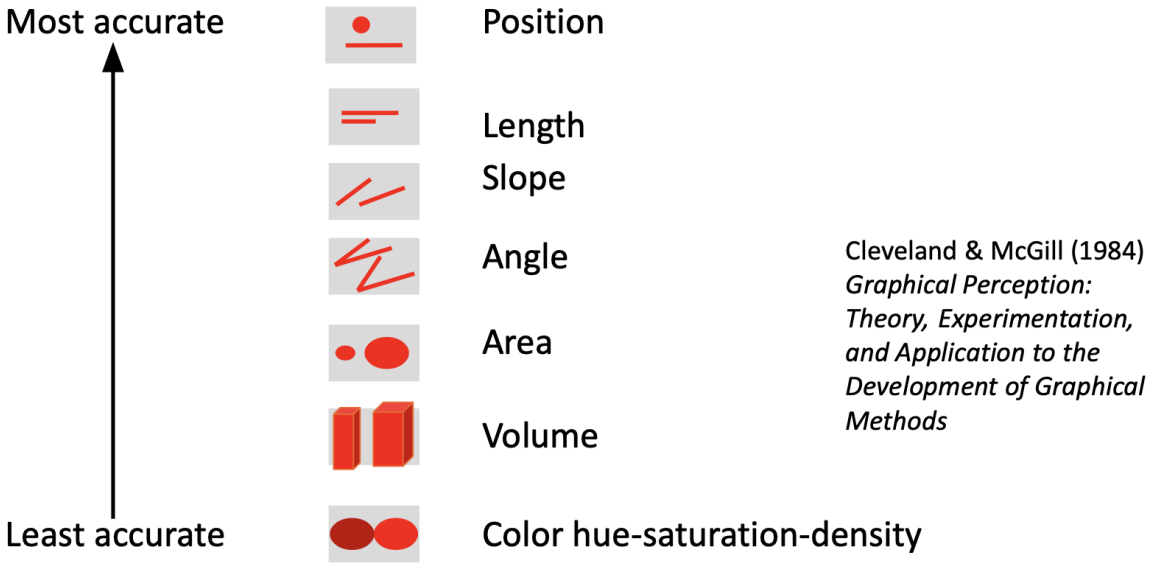
Part 2: Principles and best practices

Perception of magnitudes: some differences are difficult to see with little time.

Just noticable difference (JND)

- Weber's law: $\frac{\Delta I}{I} = k$
- I: intensity; ΔI: increase from I to notice a difference; k: constant
- Most continuous variations in stimuli are perceived in discrete (multiplicative) steps

As humans, we are most likely to underestimate comparisons between areas/surfaces



Choose your axes wisely!

linear y-axis vs. logarithmic y-axis

Visualizing heavy-tailed distributions: the median is bigger than the mean. Taking the logarithmic x-axis gives a log normal distribution. This often occurs in nature.

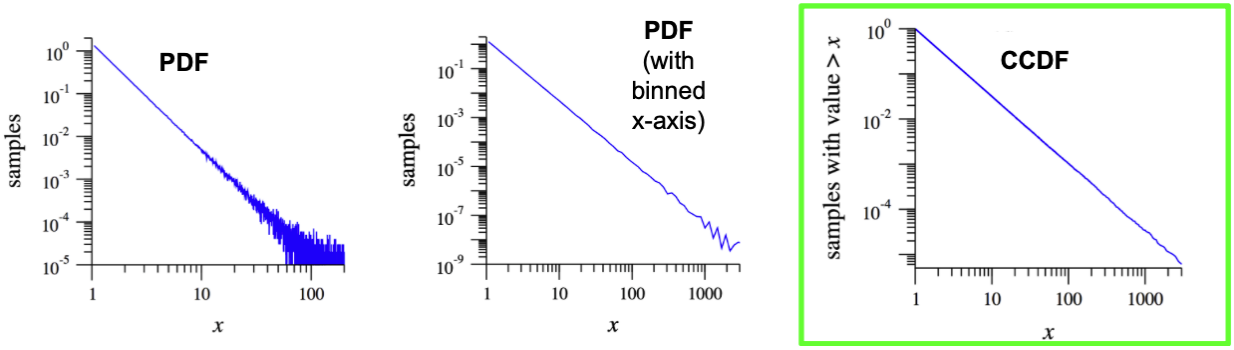
Heavy-tailed data: power laws

- Probability of x: $p(x) = Cx^{-a}$
  - Very large values are rare, "but not very rare"
  - Body size vs. city size
  - Many natural phenomena are power laws (e.g., of friends)
  - For dealing with them, need to know some tricks
  - E.g., for small a, mean var $= \infty \Rightarrow$ use median!
  - E.g., straight line on log-log axes: $y = Cx^{-a} \Leftrightarrow \log y = \log C - a \log x$

- Complementary cumulative distribution function (CCDF): $P(x) := Pr\{X \geq x\}$
- CCDF of power law is also a power law (with exponent $\alpha - 1$)

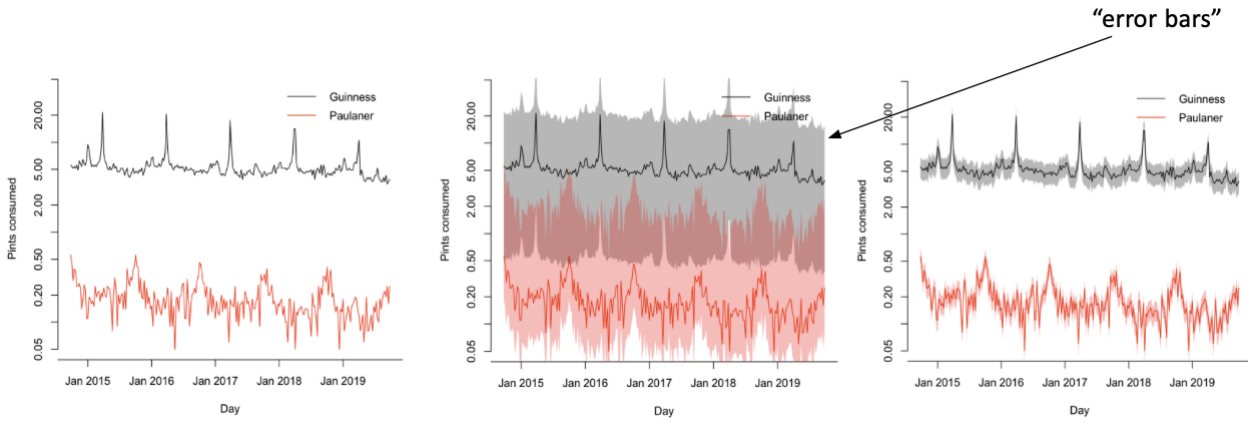
$$P(x) = C \int_x^\infty x'^{-\alpha} dx' = \frac{C}{\alpha - 1} x^{-(\alpha-1)}.$$

- CCDF plot is monotonically decreasing (even without binning)



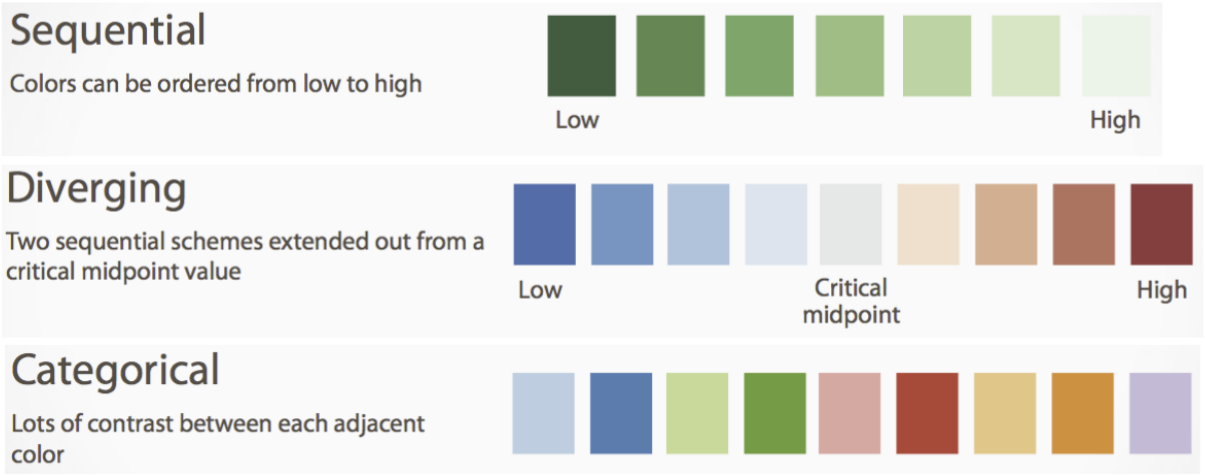
Show data uncertainty!

Which beer is more popular, **Guinness** or Paulaner ?



Use colours wisely!

Choose colours based on the information you want to convey



Use visual contrast

- Use the visual contrast to highlight the important information
- Vary the size, shape, position, orientation, or color of an element to make the key part of the figure the most visually prominent
- Avoid incorporating every form of visual contrast

Part 3: a (small) selection of use cases for data visualization

- Presenting scientific results
- Data wrangling (multimodal data)
  - Two or more distinct peaks in a histogram often **suggest 2 or more distinct populations** of samples
  - But don't guess! Explore further by using, e.g., color and a histogram of multiple populations
- Data wrangling (weird data)
  - Maintain a **theory of what the data should look like**
  - Some data is **very hard to explain**
  - Never just blink it away! First, assume a bug. Try to fix it
  - If not a bug: you might have made an interesting discovery!
  - Some of science's most important findings were made by not ignoring weird data, but dwelling on it!
- Journalism